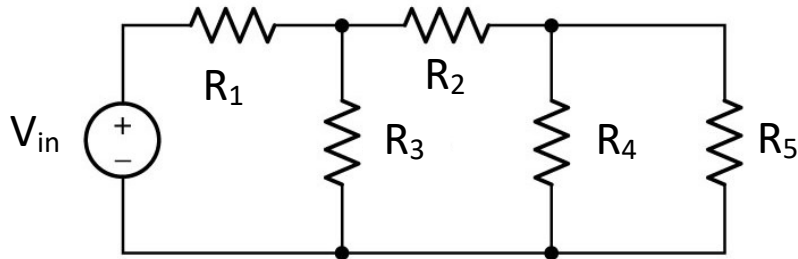


Circuit Exercises

The purpose of this experiment is to practice with simple circuits and their analysis.

Procedure:

1. Build the network shown in the Figure (use your choice of resistors). Set up the Power Supply to serve as the voltage source V_{in} . In this experiment we use $V_{in} = 6 - 9\text{ V}$.



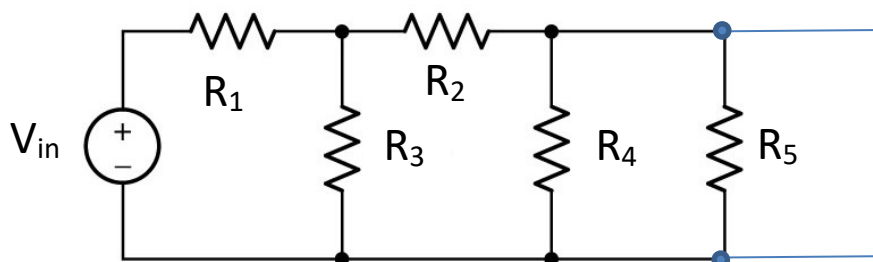
2. Calculate the voltages and currents in this circuit using voltage and current division laws.

- Measure the voltage across each of the resistors. Pay special attention to the voltage signs.
- Set up the DMM as a DC ammeter. Measure the current through each of the resistors. Keep in mind that you need to open the circuit at either terminal of the resistor of interest. The circuit is then closed through the DMM.
- Compare your experimental results with the calculated results in Table 1.

Table 1.

Resistor #	Resistance nominal	Resistance measured	Voltage calculated	Voltage measured	Current calculated	Current measured
1						
2						
3						
4						
5						

- Short the resistors R_4 and R_5 with the wire as shown in the diagram.



- Measure the voltage across each resistor. Pay special attention to the voltage signs.
- Set up the DMM as a DC ammeter. Measure the current through each resistor.
- Put your measured data in Table 2.

Table 2.

Resistor #	Voltage calculated	Voltage measured	Current calculated	Current measured
1				
2				
3				
4				
5				

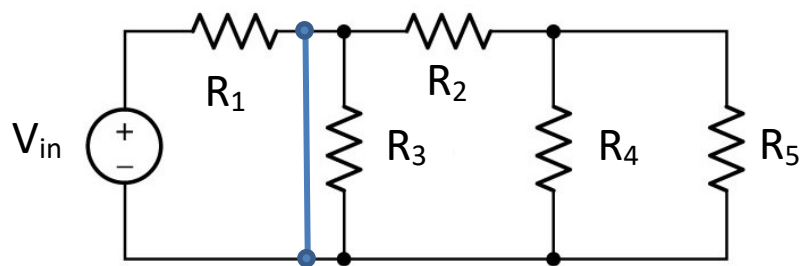
10. Are the voltages across R_1 and R_3 changed compare to Table 1? Why?

11. Are the currents through the resistors changed? Why?

12. If you now remove the resistors R_4 and R_5 (but leave a wire), are there any changes in Table 2? Why?

13. Draw an equivalent circuit and calculate voltages and currents again for this circuit. Compare these theoretical values in Table 2.

14. Now, short the resistor R_3 with the wire as shown in the diagram below.



15. Measure the voltage across each resistor. Pay special attention to the voltage signs.

16. Set up the DMM as a DC ammeter. Measure the current through each resistor.

17. Put your measured data in Table 3.

Table 3.

Resistor #	Voltage calculated	Voltage measured	Current calculated	Current measured
1				
2				
3				
4				
5				

18. What are your conclusions? Explain what happened to the currents through the resistors. Draw an equivalent circuit and calculate theoretical values in Table 3.

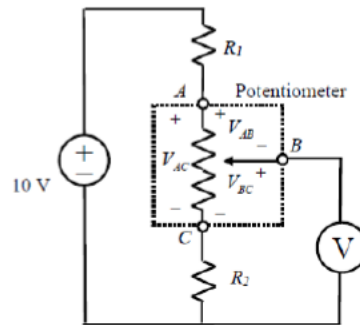
19. Replace R_2 in an initial circuit with a diode in direct bias. Measure currents through the resistors R_4 and R_5 .

20. Now, connect a diode in reverse bias. Measure currents through the resistors R_4 and R_5 .

21. Explain the role of diode in the circuit.

22. The purpose of this experiment is to familiarize you with the potentiometer and then use the potentiometer to perform voltage division.

a) Construct the network shown in the Figure. The instructor will supply an appropriate potentiometer.



b) Before connecting the power supply, use the DMM in ohmmeter mode to measure the resistance of the potentiometer between terminals A-C, A-B, and B-C. Investigate how these resistances change as the wiper pin is rotated. Record your observations below. Specifically, make sure to record the resistance between terminals A-C, as this information will be needed later.

c) When finished with your resistance measurements, set up the Power Supply to serve as the 10-V dc source.

d) Set up the DMM as a DC voltmeter and connect through terminals B-C as shown in diagram.

e) Turn the wiper of the potentiometer as far as it will go from one terminal position to another using a screwdriver or “tweak tool”.

f) Observe how voltage is changing.

g) Check voltage division law for at least 3 wiper positions. Write down your results below.